Trustworthy Fault Diagnosis Method Based on Belief Rule Base With Multisource Uncertain Information for Vehicle

Zhichao Feng, Ruohan Yang, Zhijie Zhou, and Changhua Hu

***Abstract*—This article develops a new trustworthy fault diagnosis (TFD) method based on belief rule base expert system with multisource uncertainty information (BRB-MU) for a vehicle, such as a flying machine. To improve the performance of fault diagnosis for a vehicle, a new BRB-MU is developed, where the influence of the random environment disturbance is addressed by introducing uncertain factors of input characteristics. In BRB-MU, the uncertain factors of input characteristics are calculated by the signal to noise ratio based method. The new developed BRB-MU model aims to handle the following three problems: 1) unbalanced observation data of system states; 2) uncertain expert knowledge and 3) random environment disturbance. The first two problems are solved by the combination of expert knowledge and observation data. In parallel, the third problem is handled by the new introduced uncertain factors based on singular value decomposition in BRB-MU. To quantitatively analyze the influence of the uncertain information to the TFD output, the traceability analysis of the BRB-MU is conducted that can provide support for decision making of vehicle optimization design. It is analyzed based on the modeling traceability and parameters interpretability. To demonstrate the effectiveness of the TFD method, an experiment illustration is conducted.**

***Index Terms*—Belief rule base, fault diagnosis (FD), multisource uncertain information, trustworthy, vehicle.**

I. Introduction

VEHICLES, represented by a flying machine, are important transport carriers for satellites in aerospace. With the development of electronic technology, systems, such as a rocket and a drone, are developing in the direction of complexity and integration. In engineering practice, performance maintenance of the vehicle during its operation is an important way to improve its reliability and avoid serious accidents [1], [2].

Fault diagnosis (FD) is a useful way to detect and diagnose the failure of the vehicle. It is conducted based on the gathered observation information that determines its diagnosis accuracy [3], [4], [5]. In current research studies of FD methods, they can be divided into the following three categories by the observation information they applied in the modeling process: 1) the knowledge-based method; 2) the data-based method and 3) the mixed information-based method. First, the knowledge-based method diagnoses the system failure by the information provided by experts according to their experience. Its modeling process is transparent and intervened by experts, but its modeling ability is limited by the ability of experts. For example, Qin et al. [6] developed an FD method based on Bayesian discriminating analysis for concurrent failure, and Guo et al. [7] diagnosed the failure of fault tolerance permanent magnet fault tolerant permanent magnet synchronous motor system (FTPMSM) by the average current Park's vector (ACPV) sign symbol. Second, the data-based methods are conducted based on a large amount of system observation data, and its modeling process is simple and has been widely used. However, as the main source of system information, once the observation data quality does not satisfy the requirement, the modeling ability of the data-based method is reduced. For example, support vector machines (SVM) and neural network have been applied in the FD method for rotating machinery, heat exchanger, transistor, etc. [8], [9], [10]. Third, for the mixed information-based method, it is developed based on the system knowledge and the observation data simultaneously. Hence, it has the advantages and disadvantages of the above two methods. At the same time, the multisource uncertainty information also provides challenge for the modeling ability of the mixed information-based method. For example, Chang et al. [11] and Xu et al. [12] proposed the FD methods based on belief rule base expert system for marine diesel engines.

In engineering practice, in order to improve the diagnosis accuracy of the FD for a vehicle, the following three challenges need to be solved: 1) lack of fault data; 2) uncertain knowledge; and 3) environmental disturbance. Comparing the above three categories FD methods, the knowledge-based method and the data-based method cannot solve the first and the second challenges, respectively. The mixed information-based method can

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solve the first two challenges while it cannot address the third problem.

Belief rule base expert system (BRB) is one of the mixed information-based method, and it has been proposed based on evidential reasoning algorithm (ER), fuzzy theory, and IF-THEN rule by Yang et al. [13], [14], [15]. The modeling process of BRB is conducted based on belief rules that is applied to profile the nonlinear relationship between the system input and output. The structure of belief rules is consistent with human judgement. The structure and the initial parameters are determined by experts. Hence, experts can intervene the modeling process of BRB that ensure its interpretability [16], [17], [18]. That is to say, the output of the BRB is trustworthy. Moreover, the BRB model can use the expert knowledge and the observation data simultaneously, which can increase the amount of the modeling information of a system. Hence, the BRB model can address the first two problems as analyzed above. However, in engineering practice, the working environment of a vehicle is complex and unpredictable. The gathered observation information has noise, and the knowledge provided by experts and the data obtained by sensors are both uncertain. In current studies of BRB, its inputs are assumed to be cleaned or only the reliability of the observation data has been considered [19], [20], [21]. Current BRB models cannot handle these multisource uncertainty information that decreases the modeling ability of BRB. Thus, in order to improve the diagnosis accuracy of vehicle, a new BRB model with multisource uncertainty information (BRB-MU) is developed and then the BRB-MU based trustworthy FD method (TFD) is proposed. The innovations of the article can be concluded as follows.

1) A new BRB-MU is developed. In BRB-MU, the uncertainty of the multisource information is measured by the signal to noise ratio, and it is calculated based on singular value decomposition (SVD) [22].
2) In order to demonstrate the influence of the uncertain information, the sensitivity analysis of the uncertainty factors to the model output is conducted based on the transparency of BRB-MU. It can improve the trustworthy of the output of the developed TFD method.
3) To further address the influence of the uncertainty of the expert knowledge to the initial parameters, an optimization model is constructed based on the projection covariance matrix adaption evolution strategy (P-CMA-ES). In the optimization model, the constraints of parameters are given to ensure their physical meaning in the modeling process of the BRB-MU.

The rest of this article is organized as follows. In Section II, the problems of FD for a vehicle are formulated and the BRB-MU based TFD method is constructed. Then, in Section III, the inference process of BRB-MU based TFD method is presented, where the uncertainty of multisource information is addressed by the signal to noise ratio. Section IV conducts the traceability analysis of the uncertainty to the BRB-MU output. To illustrate the effectiveness of the developed method, Section V presents an experimental illustration. Finally, Section VI concludes this article.

II. Problem Formulation and Model Construction

A. Problems Formulation of FD for a Vehicle

The problems of FD for a vehicle solved in this article can be concluded as the following aspects in this subsection.

Problem 1: Unbalanced data and inaccurate expert knowledge.

In engineering practice, a vehicle is the most important tool in aerospace. Its components are strictly screened. They have high reliability but are expensive. Thus, it is impossible to obtain enough useful data by repeating the experiment. On the other hand, the human brain has limited ability to model a complex system. Based on the observation data of the vehicle, the experts can only provide uncertain FD result. For example, when the signal of the vehicle measurement and control system is normal, but the vehicle deviates from the normal trajectory, it can be judged that the servo mechanism has a high probability of failure. Hence, in this case, the failure of the vehicle can be judged as the high failure probability of the servo mechanism. In engineering practice, the process of expert FD for a vehicle is a process of eliminating all kinds of observation data. Experts mainly rely on their own experience to diagnose faults. The obtained expert knowledge cannot be applied directly.

Problem 2: Interference of complex working environment to the gathered information.

In the flight process of the vehicle, its flight environment can be divided into the five following regions:

1) troposphere;
2) stratosphere;
3) mesosphere;
4) thermosphere;
5) exosphere.

In a different environment, the temperature, pressure, humidity, wind direction, and wind speed are different. These factors influence the flight performance of vehicles. Moreover, these factors change quickly and it is difficult to predict them accurately. Hence, these interference factors introduce uncertain information into the observation information of vehicle. These multisource uncertainty information will reduce the diagnosis accuracy of the FD method, and it should be handled in engineering practice.

The above two problems influence the diagnosis accuracy of the FD for vehicle, and they will be solved in the following developed new TFD method.

B. Construction of the BRB-MU Based TFD Method

In the BRB-MU model, the nonlinear relationship between the vehicle characteristics and diagnosis output is profiled by the belief rules, and its structure is shown in Fig. 1. There are L belief rules in BRB-MU and the kth belief rule can be represented

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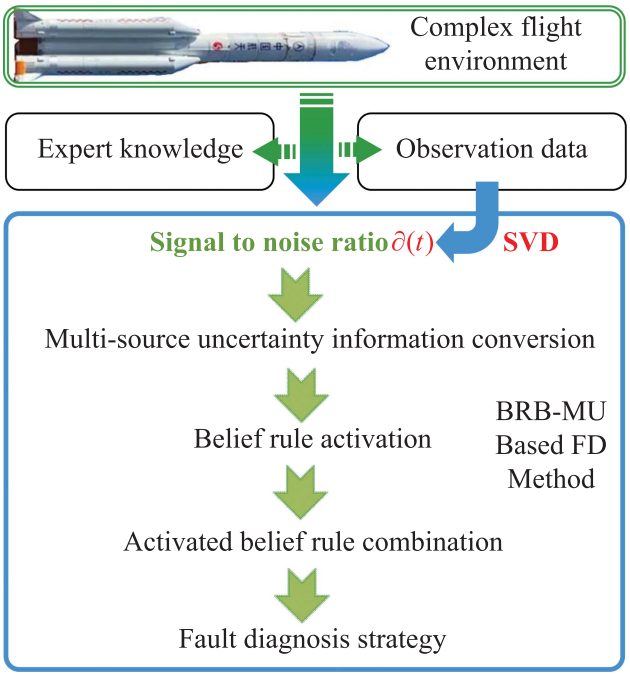


Fig. 1. Modeling process of the BRB-MU based FD method.

as [23]

$$\begin{aligned} R_k(t):&\ \text{If } C_1(t) \text{ is } A_1^k \wedge C_2(t) \text{ is } A_2^k \cdots \wedge C_M(t) \text{ is } A_M^k \\ &\text{then } F(t) \text{ is } \{(F_1, \beta_{1,k}(t)), (F_2, \beta_{2,k}(t)), \ldots \\ &(F_N, \beta_{N,k}(t)), (\Theta, \beta_{\Theta,k}(t))\} \\ &\left(\sum_{n=1}^{N} \beta_{n,k} \le 1, \sum_{n=1}^{N} \beta_n + \beta_\Theta = 1\right) \\ &\text{with rule weight } \hbar_k(t), \text{and signal to noise ratio } \partial(t) \\ &\text{characteristic weight } \delta_1(t), \cdots, \delta_M(t) \end{aligned} \tag{1}$$

where $C_1(t), C_2(t), \ldots, C_M(t)$ are the inputs of the M characteristics of vehicle. $F(t)$ denotes the diagnosis output of the BRB-MU model at time instant t. $\{F_1, F_2, \ldots, F_N\}$ represent the fault conditions of the vehicle, and $\{\beta_{1,k}(t), \beta_{2,k}(t), \ldots, \beta_{N,k}(t)\}$ are their corresponding belief degrees. Θ represents subset of all the fault conditions of the vehicle and $\beta_{\Theta,k}(t)$ is the global ignorance that cannot be determined which fault conditions should be assigned. $\hbar_k(t)$ and $\delta_m(t)$, $m = 1, \ldots, M$ denote the weight of the kth belief rule and the mth input characteristic [24], respectively. $\partial(t)$ is the signal to noise ratio that represents the influence of the working environment to the gathered multisource information. The reference points of input characteristics and the output degrees are determined by experts according to the requirement of system accuracy and real-time performance. Then, the belief degrees of output degrees in each belief rules are determined by experts by the input combination of reference points of characteristics. In the process, the expert knowledge is aggregated into the BRB-MU model. Influenced by the uncertainty of expert knowledge, the initial parameters of the BRB-MU model are not accurate. The small amount of observation data are further applied to train the parameters. By the optimization model, the expert knowledge and the observation data are fused. Note that the sample size of observation data needs to be defined according to the complexity of the system. Small sample refers to the inability to learn the behavior of the system state through the data of the existing system state.

Remark 1: Compared with traditional BRB model, the BRB-MU model further considers the influence of the multisource uncertainty information, where the signal to noise ratio is applied to represent its influence. In current studies of BRB, the researchers focus on the optimization method and its application in engineering practice, and the influence of uncertain information has not been considered. Therefore, current BRB models cannot handle the multisource uncertainty information, especially observation data. In parallel, there are also other methods to address the multisource uncertainty information, such as a filter. However, filtering also generates errors that may pass between the BRB-MU and the filter model. Hence, in this article, the signal to noise ratio $\partial(t)$ is applied in BRB-MU.

III. Inference Process of BRB-MU Based FD Method

A. Calculation Method of Signal to Noise Ratio Based on SVD

In vehicles, assume that M characteristics exist and each characteristic has T observation data. The observation data of characteristics can be profiled as

$$\mathbf{X} = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1M} \\ C_{21} & C_{22} & \cdots & C_{2M} \\ & & \ddots & \\ C_{T1} & C_{T2} & & C_{TT} \end{bmatrix}. \tag{2}$$

Then, the matrix of observation data $\mathbf{X}$ can be decomposed into the following formula:

$$\mathbf{X} = \mathbf{U} \sum \mathbf{V}^{\mathrm{T}} = \sum_{i=1}^{r} \sigma_i \mathbf{U}_i \mathbf{V}_i^{\mathrm{T}} \tag{3}$$

where $\mathbf{U}$ is the eigenvector matrix of the eigenvalue of $\mathbf{X}\mathbf{X}^{\mathrm{T}}$, and $\mathbf{V}$ is the eigenvector matrix of the eigenvalue of $\mathbf{X}^{\mathrm{T}}\mathbf{X}$. Σ denotes the diagonal matrix composed by the singular value of $\mathbf{X}$, and it can be further represented as

$$\Sigma = \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix}. \tag{4}$$

Hence, the average energy of noise and observation data can be calculated by the following formulas:

$$E_N = \frac{1}{r-1} \sum_{i=2}^{r} \sigma_i^2 \tag{5}$$

$$E_S = \sigma_1^2 - \frac{1}{r-1} \sum_{i=2}^{r} \sigma_i^2 \tag{6}$$

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where E_N and E_S are the average energy of noise and observation data, respectively.

The signal to noise ratio of the observation data of vehicle characteristics is obtained by

$$\kappa_{STR} = \frac{E_S}{rE_N}. \tag{7}$$

Remark 2: In the developed SVD-based signal to noise ratio calculation method, assume that all the sensors in a vehicle have same performance and their observation data are only influenced by the working environment. Hence, all the characteristics of the vehicle have same signal to noise ratio.

B. Inference Process of BRB-MU Based FD Method

First, once the observation data of characteristics are available, they can be transformed into a unified framework by the following formula:

$$\varpi_j^i(t) = \begin{cases} \frac{A_i^{k+1} - C_i(t)}{A_i^{k+1} - A_i^k} & j = k, \text{ if } A_i^k \le C_i(t) \le A_i^{k+1} \\ \frac{C_i(t) - A_i^k}{A_i^{k+1} - A_i^k} & j = k+1 \\ 0 & j = 1, 2, \ldots, L, j \ne k, k+1 \end{cases} \tag{8}$$

where $\varpi_j^i(t)$ is the matching degree of the ith characteristic to the jth belief rule. L denotes the number of the belief rules in BRB-MU. A_i^k and A_i^{k+1} are the reference points of the ith characteristic in the kth and $k+1$th belief rule, respectively. The reference points of characteristics are applied to translate the multisource information into the unified matching degree. $C_i(t)$ denotes the input information of the ith characteristics at time instant t.

Then, based on the structure of the belief rules in BRB-MU, different belief rules have different activation degrees that can be measured by the activation weight. The activation weight of the belief rule can be obtained by

$$w_k(t) = \frac{\theta_k(t)\varpi_k(t)}{\sum_{l=1}^{L}\theta_l(t)\varpi_l(t)}, \quad k = 1, \ldots, L \tag{9}$$

with

$$\varpi_k(t) = \prod_{i=1}^{M}\left(\left(1 - \frac{1}{\kappa_{STR}}\right)\varpi_k^i(t)\right)^{\overline{\delta}_i(t)} \tag{10}$$

$$\overline{\delta}_i(t) = \frac{\delta_i(t)}{\max\limits_{i=1,\ldots,M}\{\delta_i(t)\}}, 0 \le \overline{\delta}_i(t) \le 1 \tag{11}$$

where $w_k(t)$ is the activation weight of the kth belief rule. $\varpi_k(t)$ denotes the matching degree of characteristics' inputs to the kth belief rule. Note that the signal to noise ratio of characteristics κ_{STR} is aggregated into the matching degree $\varpi_k(t)$.

On the basis of the activation weight, the output belief degrees of belief rules can be transformed into the basic probability masses (*bpm*) by

$$P_{n,k}(t) = w_k(t)\beta_{n,k}(t), \ j = 1, \cdots N \tag{12}$$

$$P_{D,k}(t) = 1 - \sum_{n=1}^{N} P_{n,k}(t) = 1 - w_k \sum_{n=1}^{N}\beta_{n,k}(t) \tag{13}$$

$$\overline{P}_{D,k}(t) = 1 - w_k(t) \tag{14}$$

$$\widetilde{P}_{D,k}(t) = w_k(t)\left(1 - \sum_{n=1}^{N}\beta_{n,k}(t)\right) \tag{15}$$

where $P_{n,k}$ represents the *bpm* of the nth diagnosis output in the kth belief rule. $P_{D,k}(t)$ is the remaining *bpm* unassigned to any diagnosis output and $P_{D,k}(t) = \overline{P}_{D,k}(t) + \widetilde{P}_{D,k}(t)$. $\overline{P}_{D,k}(t)$ and $\widetilde{P}_{D,k}(t)$ are generated by incomplete activation and incompleteness of the kth belief rule, respectively.

The activated belief rules are aggregated by the ER algorithm as follows:

$$\begin{aligned} P_{n,er(k)}(t) = \hbar_{er(k)}(t)[&P_{n,er(k-1)}(t)P_{n,k}(t) \\ &+ P_{n,er(k-1)}(t)P_{D,k}(t) + P_{D,er(k-1)}(t)P_{n,k}(t)] \end{aligned} \tag{16}$$

$$P_{D,er(k)}(t) = \bar{P}_{D,er(k)}(t) + \tilde{P}_{D,er(k)}(t) \tag{17}$$

$$\begin{aligned} \tilde{P}_{D,er(k)}(t) = \hbar_{er(k)}(t)[&\tilde{P}_{D,er(k-1)}(t)\tilde{P}_{D,k}(t) \\ &+ \tilde{P}_{D,er(k-1)}(t)\bar{P}_{D,k}(t) + \bar{P}_{D,er(k-1)}(t)\tilde{P}_{D,k}(t)] \end{aligned} \tag{18}$$

$$\bar{P}_{D,er(k)}(t) = \hbar_{er(k)}(t)[\bar{P}_{D,er(k-1)}(t)\bar{P}_{D,k}(t)] \tag{19}$$

$$\hbar_{er(k)}(t) = \left[1 - \sum_{n=1}^{N}\sum_{\substack{t=1 \\ t\ne n}}^{N} P_{n,er(k-1)}(t)P_{t,k}(t)\right]^{-1} \tag{20}$$

where $P_{n,er(k)}(t)$ is the combined *bpm* of the nth diagnosis output of the first k belief rules. $\hbar_{er(k)}(t)$ is the normalized coefficient to ensure the constraint $\sum_{n=1}^{N} P_{n,er(L)}(t) + P_{D,er(L)}(t) = 1$.

Finally, the belief degrees of diagnosis outputs can be obtained by

$$\beta_n(t) = \frac{P_{n,er(L)}(t)}{1 - P_{D,er(L)}(t)}, \ n = 1, 2, \ldots, N \tag{21}$$

$$\beta_D(t) = \frac{P_{D,er(L)}(t)}{1 - P_{D,er(L)}(t)} \tag{22}$$

where $[\beta_1(t), \beta_2(t), \ldots, \beta_N(t)]$ denotes the belief degree vector of vehicle diagnosis outputs. $\beta_D(t)$ is the remaining belief degree of the model output. It is generated by the incompleteness of belief rules.

Then, on the basis of diagnosis output of BRB-MU, the failure of vehicle can be diagnosed by the nearest neighbor distance strategy. The distance between the BRB-MU output and the standard failure vector is calculated by

$$\|\mathbf{B}(t) - \mathbf{J}\| = \left(\sum_{n=1}^{N}|\beta_n(t) - J_n|^{\upsilon}\right)^{1/\upsilon} \tag{23}$$

where $\mathbf{B}(t) = [\beta_1(t), \beta_2(t), \ldots, \beta_N(t)]$. υ is the order of the norm. J_n is the standard of the nth failure. $\mathbf{J} = [J_1, J_2, \ldots, J_N]$. $\mathbf{J}$ denotes the standard failure vector and it is determined by

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experts. It can be represented as

$$\mathbf{J} = \begin{bmatrix} J_{11} & & J_{1N} \\ & \ddots & \\ J_{N1} & & J_{NN} \end{bmatrix} \tag{24}$$

where $J_n = [J_{n1}, J_{n2}, \ldots, J_{nN}]$.

Therefore, the failure of the vehicle can be determined by

$$F(t) = \min_{n=1,\cdots N} \|\mathbf{B}(t) - \mathbf{J}\|. \tag{25}$$

C. Optimization Model of the Developed FD Method

In order to handle the uncertainty of the expert knowledge while ensuring the physical meaning of parameters, an optimization model is developed.

To ensure the physical meaning of model parameters of BRB-MU, they should satisfy the following constraints:

$$0 \le \delta_i(t) \le 1 \tag{26}$$

$$0 \le \hbar_k(t) \le 1 \tag{27}$$

$$0 \le \beta_{n,k}(t) \le 1 \tag{28}$$

$$\sum_{n=1}^{N} \beta_{n,k}(t) \le 1. \tag{29}$$

The optimization objective of the developed optimization model is to maximize the diagnosis accuracy that can be represented as

$$D_{\text{rate}} = \frac{F}{T} \tag{30}$$

with

$$F = \sum_{t=1}^{T} F_t \tag{31}$$

where D_{rate} denotes the diagnosis rate of the FD method. F is the amount of correct diagnoses, and F_t represents the diagnosis output at the tth time instant. T is the number of observation data.

Then, the optimization model of the BRB-MU based FD method can be represented as

$$\begin{aligned} &\min \quad D_{\text{rate}} \\ &\text{s.t.} \quad (26)\text{–}(29). \end{aligned} \tag{32}$$

In this article, to train the parameters of the FD method, the P-CMA-ES algorithm is applied as the optimization algorithms, and it is one of the intelligent algorithms that can find the optimal solution of the BRB-MU based on the gathered observation data [20], [23], [24]. In BRB-MU, there is no uniform definition of physical meaning of the parameters, such as belief rule weight and belief degree. For example, the constraint of the belief rule weight is set to 0 to 1, and 1 means most important. Hence, in BRB-MU, the physical meaning of its parameters represents that these parameters can be understood by experts before and after optimization. It should be noted that, to ensure the physical meaning of the parameters, the constraints are applied for the BRB-MU parameters. The physical meaning of parameters and model structure remain unchanged before and after optimization. Hence, the transparency of the BRB-MU model can be ensured before and after optimization.

IV. Traceability Analysis of BRB-MU

To quantitatively analyze the effectiveness of the signal to noise ratio κ_{STR}, the traceability analysis of κ_{STR} to the FD output is conducted in this section.

Theorem 1: Assume that N diagnosis outputs and M characteristics exist in the BRB-MU based FD method. The influence of κ_{STR} to the the diagnose output belief degree can be represented as

$$\frac{\partial \beta_n(t)}{\partial \kappa_{STR}} = \frac{\partial \beta_n(t)}{\partial \varpi_k(t)} \frac{\partial \varpi_k(t)}{\partial \kappa_{STR}} \tag{33}$$

$$\frac{\partial \beta_D(t)}{\partial \kappa_{STR}} = \frac{\partial \beta_D(t)}{\partial \varpi_k(t)} \frac{\partial \varpi_k(t)}{\partial \kappa_{STR}} \tag{34}$$

where $\partial\beta_n(t)/\partial\kappa_{STR}$ and $\partial\beta_D(t)/\partial\kappa_{STR}$ are the influence factors of κ_{STR} to $\beta_n(t)$ and $\beta_D(t)$, respectively.

Proof of Theorem 1: $\partial\beta_n(t)/\partial\varpi_k(t)$ is the influence factor of output belief degree to the matching degree and it can be calculated by

$$\begin{aligned} &\begin{bmatrix} \frac{\partial \beta_n(t)}{\partial \varpi_1(t)} & \cdots & \frac{\partial \beta_n(t)}{\partial \varpi_L(t)} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\partial \beta_n(t)}{\partial P_{n,er(L)}(t)} & \frac{\partial \beta_n(t)}{\partial P_{D,er(L)}(t)} \end{bmatrix} \\ &\quad \times \begin{bmatrix} \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_1(t)} & \cdots & \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_L(t)} \\ \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_1(t)} & \cdots & \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_L(t)} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{1-P_{D,er(L)}(t)} & \frac{P_{n,er(L)}(t)}{1-P_{D,er(L)}(t)} \end{bmatrix} \\ &\quad \times \begin{bmatrix} \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_1(t)} & \cdots & \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_L(t)} \\ \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_1(t)} & \cdots & \frac{\partial P_{n,er(L)}(t)}{\partial \varpi_L(t)} \end{bmatrix}. \end{aligned} \tag{35}$$

Then, the influence factors of $\varpi_k(t)$ to $P_{D,er(q)}(t)$ and $P_{n,er(q)}(t)$ can be obtained by

$$\frac{\partial P_{D,er(q)}(t)}{\partial \varpi_k(t)} = \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \varpi_k(t)} + \frac{\partial \tilde{P}_{D,er(q)}(t)}{\partial \varpi_k(t)} \tag{36}$$

$$\frac{\partial P_{n,er(q)}(t)}{\partial \varpi_k(t)} = \begin{bmatrix} \frac{\partial P_{n,er(q)}(t)}{\partial \hbar_{er(q)}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{n,er(q-1)}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{n,q}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{D,q}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{D,er(q-1)}(t)} \end{bmatrix}^T \begin{bmatrix} \frac{\partial \hbar_{er(q)}(t)}{\partial \varpi_k(t)} \\ \frac{\partial P_{n,er(q-1)}(t)}{\partial \varpi_k(t)} \\ \frac{\partial P_{n,q}(t)}{\partial \varpi_k(t)} \\ \frac{\partial P_{D,q}(t)}{\partial \varpi_k(t)} \\ \frac{\partial P_{D,er(q-1)}(t)}{\partial \varpi_k(t)} \end{bmatrix} \tag{37}$$

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with

$$\begin{bmatrix} \frac{\partial P_{n,er(q)}(t)}{\partial \hbar_{er(q)}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{n,er(q-1)}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{n,q}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{D,q}(t)} \\ \frac{\partial P_{n,er(q)}(t)}{\partial P_{D,er(q-1)}(t)} \end{bmatrix} = \begin{bmatrix} P_{n,er(q-1)}P_{n,q} + P_{n,er(q-1)}P_{D,q} \\ + P_{D,er(q-1)}P_{n,q} \\ \hbar_{er(q)}(t)\left(P_{n,q} + P_{D,q}\right) \\ \hbar_{er(q)}(t)\left(P_{n,er(q-1)} + P_{D,er(q-1)}\right) \\ \hbar_{er(q)}(t)P_{n,er(q-1)} \\ \hbar_{er(q)}(t)P_{D,er(q-1)} \end{bmatrix} \tag{38}$$

where $\frac{\partial P_{n,er(q-1)}(t)}{\partial \varpi_k(t)}$ and $\frac{\partial P_{D,er(q-1)}(t)}{\partial \varpi_k(t)}$ can be further obtained by

$$\frac{\partial P_{n,er(q-1)}(t)}{\partial \varpi_k(t)} = \left.\frac{\partial P_{n,er(q)}(t)}{\partial \varpi_k(t)}\right|_{q=q-1} \tag{39}$$

$$\frac{\partial P_{D,er(q-1)}(t)}{\partial \varpi_k(t)} = \left.\frac{\partial P_{D,er(q-1)}(t)}{\partial \varpi_k(t)}\right|_{q=q-1}. \tag{40}$$

The influence of the matching degree to $\partial P_{D,er(q-1)}(t)$ in (36) can be represented as

$$\frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \varpi_k(t)} = \begin{bmatrix} \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \hbar_{er(q)}(t)} & \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \bar{P}_{D,er(q-1)}(t)} & \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \bar{P}_{D,q}(t)} \end{bmatrix} \times \begin{bmatrix} \frac{\partial \hbar_{er(q)}(t)}{\partial \varpi_k(t)} \\ \frac{\partial \bar{P}_{D,er(q-1)}(t)}{\partial \varpi_k(t)} \\ \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \bar{P}_{D,q}(t)} \end{bmatrix} \tag{41}$$

with

$$\begin{bmatrix} \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \hbar_{er(q)}(t)} & \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \bar{P}_{D,er(q-1)}(t)} & \frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \bar{P}_{D,q}(t)} \end{bmatrix} = \begin{bmatrix} \bar{P}_{D,er(q-1)}\bar{P}_{D,q} \\ \hbar_{er(q)}(t)\bar{P}_{D,q} \\ \hbar_{er(q)}(t)\bar{P}_{D,er(q-1)} \end{bmatrix} \tag{42}$$

$$\frac{\partial \bar{P}_{D,er(q-1)}(t)}{\partial \varpi_k(t)} = \left.\frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \varpi_k(t)}\right|_{q=q-1}. \tag{43}$$

In (36), $\partial \tilde{P}_{D,er(q)}(t)$ is obtained by the following formula:

$$\frac{\partial \tilde{P}_{D,er(q)}(t)}{\partial \varpi_k(t)} = \begin{bmatrix} \tilde{P}_{D,er(q-1)}\tilde{P}_{D,q} + \tilde{P}_{D,er(q-1)}\bar{P}_{D,q} \\ + \bar{P}_{D,er(q-1)}\tilde{P}_{D,q} \\ \hbar_{er(q)}(t)\left(\tilde{P}_{D,q} + \bar{P}_{D,q}\right) \\ \hbar_{er(q)}(t)\left(\tilde{P}_{D,er(q-1)} + \bar{P}_{D,er(q-1)}\right) \\ \hbar_{er(q)}(t)\tilde{P}_{D,er(q-1)} \\ \hbar_{er(q)}(t)\tilde{P}_{D,q} \end{bmatrix}^T \times \begin{bmatrix} \frac{\partial \hbar_{er(q)}(t)}{\partial \varpi_k(t)} & \frac{\partial \tilde{P}_{D,er(q-1)}(t)}{\partial \varpi_k(t)} & \frac{\partial \tilde{P}_{D,q}(t)}{\partial \varpi_k(t)} & \frac{\partial \bar{P}_{D,q}(t)}{\partial \varpi_k(t)} & \frac{\partial \bar{P}_{D,er(q-1)}(t)}{\partial \varpi_k(t)} \end{bmatrix} \tag{44}$$

with

$$\frac{\partial \tilde{P}_{D,er(q-1)}(t)}{\partial \varpi_k(t)} = \left.\frac{\partial \tilde{P}_{D,er(q)}(t)}{\partial \varpi_k(t)}\right|_{q=q-1} \tag{45}$$

$$\frac{\partial \bar{P}_{D,er(q-1)}(t)}{\partial \varpi_k(t)} = \left.\frac{\partial \bar{P}_{D,er(q)}(t)}{\partial \varpi_k(t)}\right|_{q=q-1}. \tag{46}$$

The influence factor $\frac{\partial \hbar_{er(q)}(t)}{\partial \varpi_k(t)}$ in (37), (41), and (44) can be calculated by

$$\frac{\partial \hbar_{er(q)}(t)}{\partial \varpi_k(t)} = \begin{bmatrix} \frac{\sum_{t=1,t\neq n}^{N} P_{t,q}}{\left[1 - \sum_{n=1}^{N}\sum_{t=1,t\neq n}^{N} P_{n,er(q-1)}P_{t,q}\right]^2} \\ \frac{\sum_{n=1}^{N} P_{n,er(q-1)}}{\left[1 - \sum_{n=1}^{N}\sum_{t=1,t\neq n}^{N} P_{n,er(q-1)}P_{t,q}\right]^2} \end{bmatrix}^T \begin{bmatrix} \frac{\partial P_{n,er(q-1)}(t)}{\partial \varpi_k(t)} \\ \frac{\partial P_{t,q}(t)}{\partial \varpi_k(t)} \end{bmatrix} \tag{47}$$

with

$$\frac{\partial P_{n,er(q-1)}(t)}{\partial \varpi_k(t)} = \left.\frac{\partial P_{n,er(q-1)}(t)}{\partial \varpi_k(t)}\right|_{q=q-1}. \tag{48}$$

The influence factors of the matching degree $\varpi_k(t)$ to *bpm*s $\partial P_{n,q}(t)$, $\partial P_{D,q}(t)$, $\partial \bar{P}_{D,q}(t)$, and $\partial \tilde{P}_{D,q}(t)$ are calculated as follows:

$$\frac{\partial P_{n,q}(t)}{\partial \varpi_k(t)} = \beta_{n,q}\frac{\partial w_q(t)}{\partial \varpi_k(t)} \tag{49}$$

$$\frac{\partial P_{D,q}(t)}{\partial \varpi_k(t)} = -\frac{\partial w_q(t)}{\partial \varpi_k(t)}\sum_{n=1}^{N}\beta_{n,q} \tag{50}$$

$$\frac{\partial \bar{P}_{D,q}(t)}{\partial \varpi_k(t)} = -\frac{\partial w_q(t)}{\partial \varpi_k(t)} \tag{51}$$

$$\frac{\partial \tilde{P}_{D,q}(t)}{\partial \varpi_k(t)} = \frac{\partial w_q(t)}{\partial \varpi_k(t)}\left(1 - \sum_{n=1}^{N}\beta_{n,q}\right). \tag{52}$$

The influence factor of the matching degree $\varpi_k(t)$ to the activation weight $w_q(t)$ is represented as

$$\frac{\partial w_q(t)}{\partial \varpi_k(t)} = \begin{cases} \frac{\theta_k}{\sum_{l=1}^{L}\theta_l\varpi_l}, q = k \\ \frac{\theta_q\varpi_q\theta_k}{\left(\sum_{l=1}^{L}\theta_l\varpi_l\right)^2}, q \neq k. \end{cases} \tag{53}$$

In a similar way, the influence factor of matching degree to the remaining output belief degree is obtained by the following formula:

$$\frac{\partial \beta_D(t)}{\partial \varpi_k(t)} = \frac{1}{\left[1 - P_{D,er(L)}\right]^2}\frac{\partial P_{D,er(L)}(t)}{\partial \varpi_k(t)} \tag{54}$$

where $\frac{\partial P_{D,er(L)}(t)}{\partial \varpi_k(t)}$ is calculated by (36).

Finally, $\frac{\partial \varpi_q(t)}{\partial \kappa_{STR}}$ in (33) and (34) can be calculated by

$$\frac{\partial \varpi_q(t)}{\partial \kappa_{STR}} = \frac{1}{{\kappa_{STR}}^2 - \kappa_{STR}}\sum_{i=1}^{M}\bar{\delta}_i\prod_{j=1}^{M}\left[\left(1 - \frac{1}{\kappa_{STR}}\right)\varpi_k^j\right]^{\bar{\delta}_i}. \tag{55}$$

Based on the above analysis, the Theorem 1 can be proved.

V. Experiment Illustration

A. Experiment Settings

Aerospace relay is one of the important parts in a vehicle that is applied to convert electrical signals into mechanical signals. Once it fails, the whole circuit does not work properly. With

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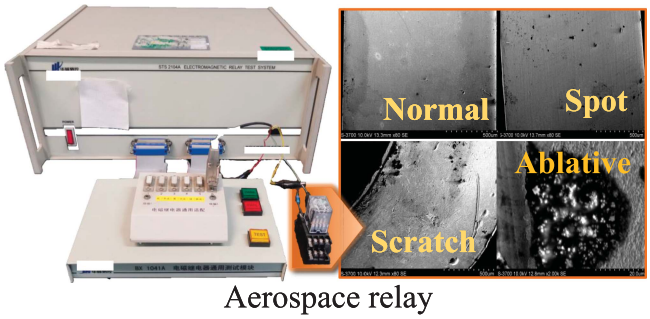


Fig. 2. Aerospace relay experiment platform.

the increase of the service time of relay, the ablation of relay contacts is serious, and its contact effect is reduced that leads to the failure. For the FD of aerospace relay, the following three problems exist: 1) unbalanced observation data; 2) uncertain expert knowledge and 3) environment disturbance. Firstly, the reliability of the aerospace relay is high. Few faults occur in the using process of aerospace relay. Hence, few complete fault samples can be obtained. This results in more normal state data and less fault state data of aerospace relay. Secondly, aerospace relays are small in size. To see the cause of the relay failure, the relay needs to be disassembled and then examined through the scanning electron microscope. There is uncertainty in the diagnostic knowledge given by experts through experience. Thirdly, the contact of aerospace relay is small in volume and light in weight. The performance of aerospace relay will be affected by temperature change and vibration in working environment. Hence, in the flight process of vehicle, there are multiple uncertainties in the information of aerospace relay obtained from by sensors.

To solve these problems of relay, the TFD model is constructed based on the developed BRB-MU model.

In the experiment, the STS 2104 A relay test platform is applied, and the JZX-22F aerospace relay is selected as the experiment subject. In parallel, to illustrate the ablative degree of relay contacts under different fault degrees, S-3700 N scanning electron microscope is used to detect the contacts. The experiment platform and the relay contacts with different fault degrees are shown in Fig. 2. It can be seen from the contacts, the degrees of electric ablation is different under different fault degrees. In the experiment, four fault degrees are selected and named as normal (N), spot (S), scratch (SC), and ablative (A). The normal state of relay denotes that the contact of the relay has not been electrically ablated, and its performance is in the best state. The spot state, scratch state, and the ablative state represent three different fault states. The ablative degree of relay contacts is increased for the three fault states. The higher the degree of ablation, the worse the relay performance. Release time (RT) and release voltage (RV) are selected as two characteristics of the relay. In the experiment, 800 sets of observation data of two characteristics are gathered, where 200 are in normal state, 200 are in spot state, 200 are in scratch state, and 200 are in ablative state.

TABLE I
REFERENCE POINTS AND VALUES OF RT

Reference point	H	SH	M	SL	L
RT	1.66	1.705	1.765	1.81	1.885

TABLE II
REFERENCE POINTS AND VALUES OF RV

Reference point	H	SH	M	SL	L
RT	1.20	1.3	1.36	1.45	1.525

TABLE III
INITIAL TFD MODEL FOR JZX-22F RELAY

No.	Inputs		Rule weight	Output fault belief degrees of aerospace relay
	RT	RV		
1	H	H	1	{0 0.3 0.3 0.3}
2	H	SH	1	{0 0.4 0.4 0.1}
3	H	M	1	{0 0.4 0.5 0.1}
4	H	SL	1	{0 0.5 0.5 0}
5	H	L	1	{0 0.5 0 0.5}
6	SH	H	1	{0 0.8 0.1 0.1}
7	SH	SH	1	{0 0.8 0.1 0.1}
8	SH	M	1	{0 0.6 0.2 0.2}
9	SH	SL	1	{0 0.5 0 0.5}
10	SH	L	1	{0 0.6 0.2 0.2}
11	M	H	1	{0 0.8 0.1 0.1}
12	M	SH	1	{0 0.8 0.2 0}
13	M	M	1	{0 0.5 0 0.5}
14	M	SL	1	{0 0.8 0 0.2}
15	M	L	1	{0 0.8 0 0.2}
16	SL	H	1	{0 1 0 0}
17	SL	SH	1	{0 0 0.5 0.5}
18	SL	M	1	{0 0 1 0}
19	SL	SL	1	{0 0 1 0}
20	SL	L	1	{0 0 0.5 0.5}
21	L	H	1	{0 0 0 1}
22	L	SH	1	{0 0.3 0.3 0.3}
23	L	M	1	{0 0 0 1}
24	L	SL	1	{0 0 0 1}
25	L	L	1	{0 0 0.3 0.7}

B. Construction of the TFD Model

The belief rules in the TFD model can be shown as (1). In the initial TFD model, the output belief degrees of belief rules, characteristic weights, and belief rule weights are given by experts and then optimized by the developed optimization model. In the experiment, five reference points are selected for the two characteristics as shown in Tables I and II, and they are high (H), slightly high (SH), medium (M), slightly low (SL), and low (L). It should be noted that the reference points must cover the fluctuation range of RT and RV. As profiled by (1), 25 belief rules exist in the BRB-MU model, and the initial TFD model is shown in Table III. Based on the calculation method as shown in Section III-A, the signal to noise ratio of RT and RV are 0.9135 and 0.8808, respectively.

C. Training and Testing of TFD

In the experiment, 800 sets of observation data of RT and RV are applied, where 400 sets are selected as the training data and the remaining data are used as testing data. The iteration

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TABLE IV
OPTIMIZED TFD MODEL FOR JZX-22F RELAY

No.	Inputs		Rule	Fault belief degrees
	RT	*RV*	weight	of relay
1	*H*	*H*	0.2755	{0.3167 0.2066 0.3264 0.1503}
2	*H*	*SH*	0.9555	{0.4752 0.1682 0.2006 0.1560}
3	*H*	*M*	0.3525	{0.1358 0.5607 0.4092 0.0000}
4	*H*	*SL*	0.6142	{0.2998 0.0000 0.0472 0.6981}
5	*H*	*L*	0.5087	{0.0971 0.2389 0.1488 0.5151}
6	*SH*	*H*	0.5923	{0.3775 0.4332 0.0259 0.1634}
7	*SH*	*SH*	0.6715	{0.4608 0.1597 0.0871 0.2924}
8	*SH*	*M*	0.0010	{0.4007 0.2581 0.1285 0.2127}
9	*SH*	*SL*	0.7387	{0.0181 0.2107 0.4440 0.3271}
10	*SH*	*L*	0.1559	{0.3465 0.2658 0.2453 0.1424}
11	*M*	*H*	0.1540	{0.0000 0.5954 0.0000 0.4553}
12	*M*	*SH*	1.0000	{0.0348 0.6945 0.2749 0.0000}
13	*M*	*M*	0.6054	{0.3293 0.2648 0.4533 0.0000}
14	*M*	*SL*	0.0735	{0.2690 0.1159 0.3113 0.3038}
15	*M*	*L*	0.2763	{0.0951 0.3212 0.4128 0.1709}
16	*SL*	*H*	0.7280	{0.4659 0.5971 0.0421 0.0000}
17	*SL*	*SH*	0.4550	{0.4086 0.1052 0.0114 0.4749}
18	*SL*	*M*	0.0462	{0.4869 0.2621 0.1639 0.0871}
19	*SL*	*SL*	0.6478	{0.0504 0.2558 0.2376 0.4562}
20	*SL*	*L*	0.1281	{0.4506 0.2904 0.2658 0.0000}
21	*L*	*H*	0.4431	{0.1223 0.4020 0.3301 0.1456}
22	*L*	*SH*	0.5375	{0.2420 0.0007 0.3223 0.4349}
23	*L*	*M*	0.3803	{0.4355 0.0917 0.1666 0.3062}
24	*L*	*SL*	0.7263	{0.0532 0.2815 0.0615 0.6038}
25	*L*	*L*	0.5548	{0.1058 0.0000 0.6704 0.2614}

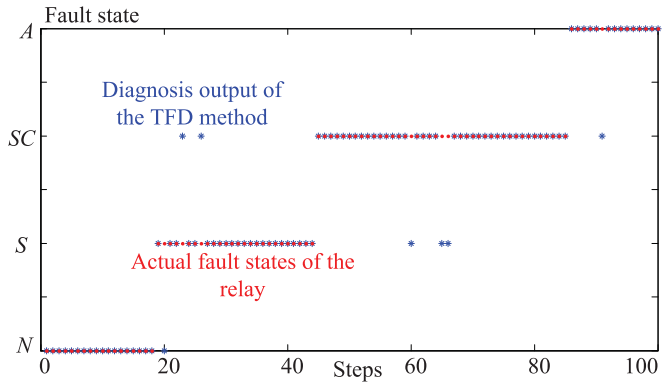


Fig. 3. Diagnosis output of the optimized TFD model.

number of P-CMA-ES is set to 500. The optimized TFD model can be shown in Table IV, and the diagnosis output is shown in Fig. 3. The diagnosis accuracy is 93.50%. It can be seen from Fig. 3 that the optimized TFD model can diagnose the aerospace relay accurately compared with the initial TFD model. To further demonstrate the robustness of the developed TFD model, the experiment is conducted for 50 times, and the diagnosis outputs are shown in Fig. 4. The mean and variance of diagnosis accuracy are 89.02% and 7.7751E-04, respectively. The variance is far smaller than the mean, and it can illustrate the robustness of the developed TFD model. In addition, the distributions of the diagnosis accuracy in four fault states can be shown in Fig. 5. It can be seen that the TFD method has the highest diagnosis performance under normal state of the aerospace relay. In spot, scratch, and ablative fault states of the relay, the fault characteristics have certain correlation that influence the diagnosis performance of the TFD method.

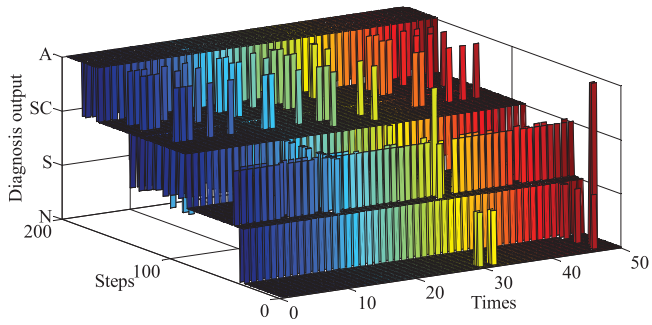


Fig. 4. Diagnosis output of the TFD model for 50 times.

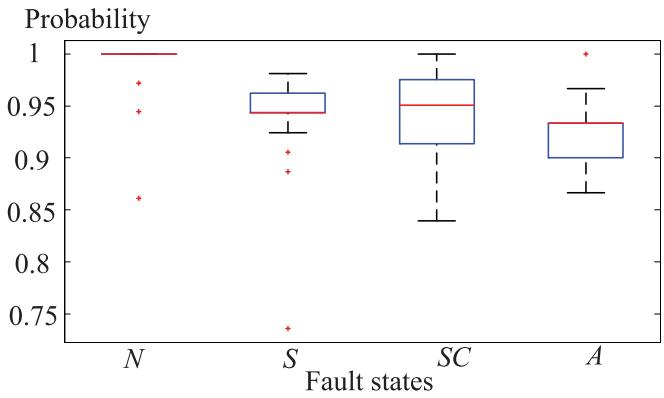


Fig. 5. Diagnosis output under four fault states of TFD.

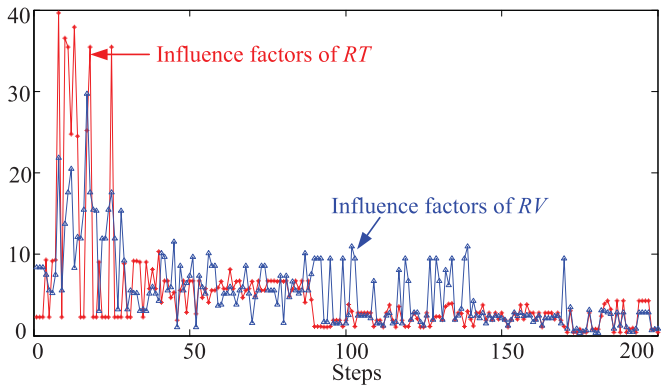


Fig. 6. Influence factors of RT and RV.

D. Traceability Analysis and Comparative Experiments

To illustrate the influence of the signal to noise ratio in the diagnosis output of aerospace relay, the traceability analysis is conducted based on Theorem 1. The influence factors of RT and RV in the experiment can be shown in Fig. 6. It can be seen that the RT has higher effectiveness to the diagnosis output. That is to say, the accuracy improvement in RT can improve the diagnosis accuracy more effectively.

Moreover, to further demonstrate the effectiveness of the developed TFD method, comparative experiments are conducted with neural network, fuzzy theory, and BRB models proposed by Zhou et al. [20], Cao et al. [23], Li et al. [24], Chang

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TABLE V
COMPARATIVE EXPERIMENTS

No.	Methods	Diagnosis accuracy
1	BRB-MU	93.50%
2	Neural network	73.00%
3	Fuzzy theory	70.00%
4	BRB model of Zhou et al.	82.60%
5	BRB model of Chang et al.	81.50%
6	BRB model of Li et al.	83.50%
7	BRB model of Cao et al.	83.65%

et al. [25]. In neural network, its maximum number of trainings, training requirement accuracy, learning rate, maximum number of failure, and minimum gradient requirement are set to 4000, 5E-10, 0.01, 5, and 1E-10, respectively. The parameters of fuzzy theory, initial parameters of BRB models of Zhouet al., Chang et al., Li et al., and Cao et al. are given by same experts. The training data and testing data of the BRB models of Zhou et al., Chang et al., Li et al., and Cao et al. are same with the BRB-MU model. The diagnosis accuracy of the comparative methods can be shown in Table V. In comparative studies, the neural network and the fuzzy theory are the typical methods of the data-based method and the knowledge-based method, respectively. In the experiment, the available data are scarce and unreliable while the expert knowledge is uncertain. These factors both influence the modeling ability of the neural network and the fuzzy theory. It can be seen that the diagnosis accuracy of BRB-MU improves 28.08% and 33.57% compared with the neural network and fuzzy theory, respectively. On the other hand, compared with the BRB models proposed by Zhou et al., Chang et al., Li et al., and Cao et al., the diagnosis accuracy of BRB-MU improves 13.20%, 14.72%, 11.98%, and 11.78%, respectively. Hence, we can obtain a conclusion that the developed BRB-MU model can handle the influence of the complex environment more effectively.

VI. CONCLUSION

This article developed a new TFD method for a vehicle that aims to solve the following three problems in engineering practice: 1) less observation data under system failure; 2) limited cognitive ability of experts; and 3) complex environment interference.

Two innovations exist in this article. First, a new BRB-MU is developed, where the uncertain factor obtained by SVD is applied. The BRB-MU model is developed based on the IF-THEN rules that are constructed by experts and then trained by observation data. The modeling process of BRB-MU is interpretability and transparency, which can guarantee the trustworthy of the TFD method. On the other hand, the influence of uncertain information is analyzed quantitatively based on the transparency of the BRB-MU. When resources are limited, to improve the diagnostic accuracy of a fault state, system improvement should be carried out according to the size of influence factors of the signal to noise ratio. Thus, the obtained influence factor of uncertain information can provide reference for the TFD method in the practical system to improve vehicle design.

In this article, as one of the intelligent optimization algorithms, the P-CMA-ES algorithm may lead to reduced model traceability. Some new methods based on BRB-MU characteristics need to be researched. Moreover, the aerospace relay is selected as the experiment subject. The aerospace relay is the basic electronic component of the vehicle, and is used in large quantity. Aerospace relay failure is one of the key reasons for vehicle control system failure. However, in vehicle, inertial navigation system, servo mechanism and so on are more important than relay. They have a different failure mechanism than relays. These challenges need to be studied in our further works.

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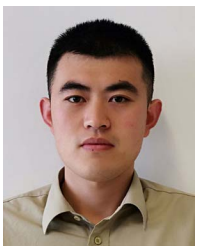
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